A_Data_Driven_Analysis_of_Phishing_Website_Characteristics_using_Python

Final Project Documentation | Dataset: PhiUSIIL Phishing URL Dataset

1. Project Overview

This project analyzes phishing and legitimate websites using Python to identify patterns in URL structures, domain characteristics, security indicators and website features. The work includes data cleaning, exploratory data analysis, statistical comparisons, visualizations and Logistic Regression classification.

2. Data Source

- Dataset Name: PhiUSIIL Phishing URL Dataset
- Source: UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset)
- Format: CSV
- Sample records analyzed: 49,999
- Original columns: 56 (including FILENAME); after dropping FILENAME and adding URLLengthCategory: 56 columns
- Target label: 0 = phishing, 1 = legitimate

3. Problem Statement

Phishing websites can imitate legitimate websites and mislead users. This analysis investigates which URL, domain and website attributes are associated with phishing and evaluates whether a classification model can distinguish the two groups within the sampled dataset.

4. Objectives

The main objective of this project is to perform Exploratory Data Analysis (EDA) on the PhiUSIIL Phishing URL Dataset using Python to identify patterns and characteristics associated with phishing and legitimate websites.

The specific objectives are:

1. To investigate URL and domain length patterns between phishing and legitimate websites.
2. To analyze URL obfuscation, digit usage, and special characters to identify differences between phishing and legitimate URLs.
3. To examine the relationship between subdomain counts and website classification.
4. To compare URL similarity scores and domain-title matching scores across phishing and legitimate websites.
5. To analyze website features such as titles, favicons, descriptions, and password fields across phishing and legitimate websites.
6. To examine HTTPS usage and other URL-related security indicators across phishing and legitimate websites.

5. Attribute (Column / Feature) Details

Attribute	Description
URLLength	Length of the URL
DomainLength	Length of the domain
IsDomainIP	Whether the domain uses an IP address
TLD	Top-level domain
TLDLegitimateProb	Estimated legitimacy probability associated with the TLD
URLSimilarityIndex	URL similarity score
NoOfSubDomain	Number of subdomains
NoOfDegitsInURL	Count of digits in URL (dataset spelling)
NoOfOtherSpecialCharsInURL	Count of special characters in URL
IsHTTPS	Whether HTTPS is used
DomainTitleMatchScore	Domain-title matching score
URLTitleMatchScore	URL-title matching score
HasTitle	Whether website title exists
HasFavicon	Whether favicon exists
HasDescription	Whether description exists
NoOfURLRedirect	Number of URL redirects
URLLengthCategory	Engineered Short/Medium/Long URL category
label	Target: phishing (0), legitimate (1)

The table highlights key analyzed attributes; the dataset contains additional numerical and website-feature columns.

6. Tools & Technologies

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn (StandardScaler, Logistic Regression, evaluation metrics)
- Google Colab

7. Data Pre-Processing and Feature Engineering

- Checked data types, missing values, duplicate records, empty strings and numerical ranges.
- Observed 0 missing values and 0 duplicate rows; no missing-value imputation or duplicate deletion was needed.
- Reviewed potential outliers using the IQR method; outliers were not automatically removed.
- Dropped FILENAME because it was a unique record identifier.
- Created URLLengthCategory: Short (≤25), Medium (26–50), Long (>50).
- Final validation: 49,999 rows, 56 columns, no missing values or duplicate rows.

8. Analysis and Visualizations

The notebook combines exploratory visualizations and grouped statistics to compare website categories. The following charts are extracted directly from the saved outputs in the uploaded notebook.

9. Exploratory Data Analysis (EDA)

- Dataset structure and numerical/categorical summaries were inspected.
- Phishing websites: 20,625 (41.25%); legitimate websites: 29,374 (58.75%).
- HTTPS, URL length, digits in URLs, title matching and website features were examined.

10. Statistical Analysis

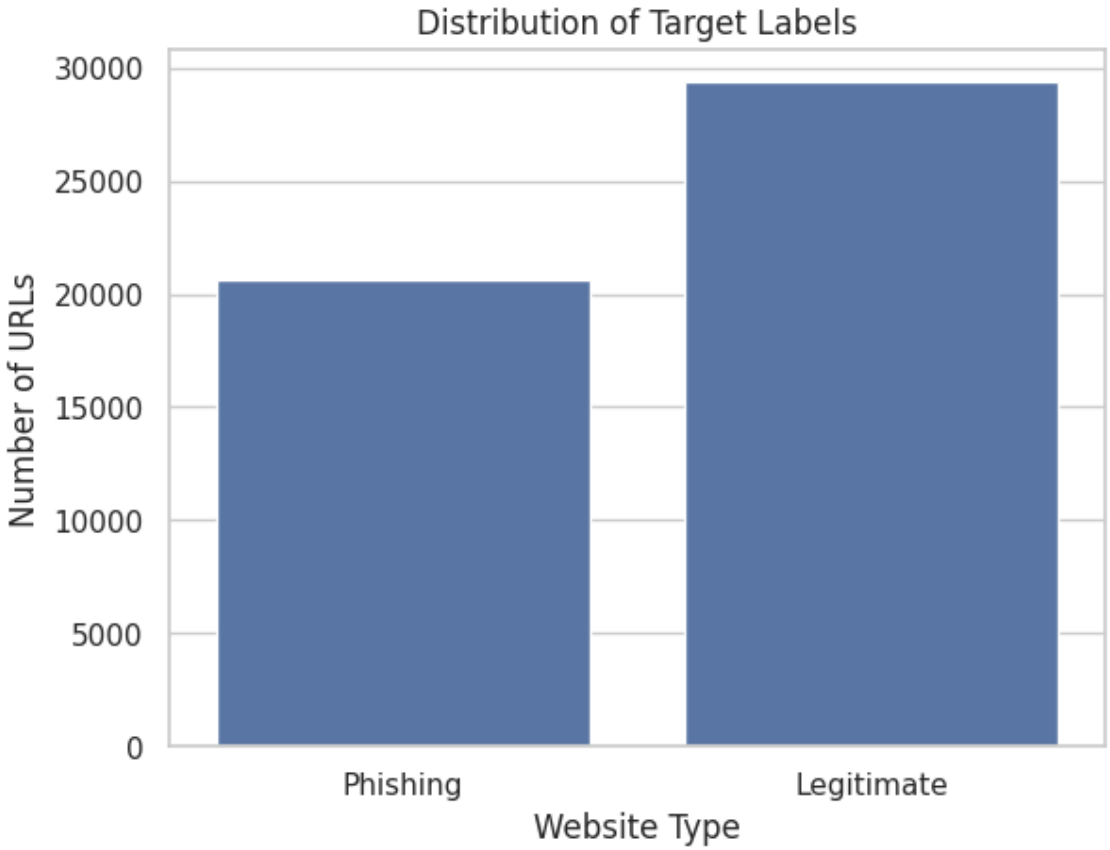
Group-wise means and distributions were calculated for URL length, domain length, special-character counts, subdomains, similarity scores, website features and security indicators.

Measure	Phishing (0)	Legitimate (1)
Average URL length	42.14	26.28
Average domain length	23.70	19.28
Average special characters	3.43	1.24
DomainTitleMatchScore mean	19.76	75.44
URLTitleMatchScore mean	23.71	75.44
HasTitle (%)	64.80	99.87
HasFavicon (%)	7.25	56.96
HasDescription (%)	4.22	73.65

11. Data Visualization

11.1 Univariate Analysis

Objective: To quantify phishing and legitimate website representation.



Target label distribution

Interpretation: The dataset contains 20,625 phishing websites (41.25%) and 29,374 legitimate websites (58.75%).

Legitimate websites are more common than phishing websites, indicating a moderate class imbalance.

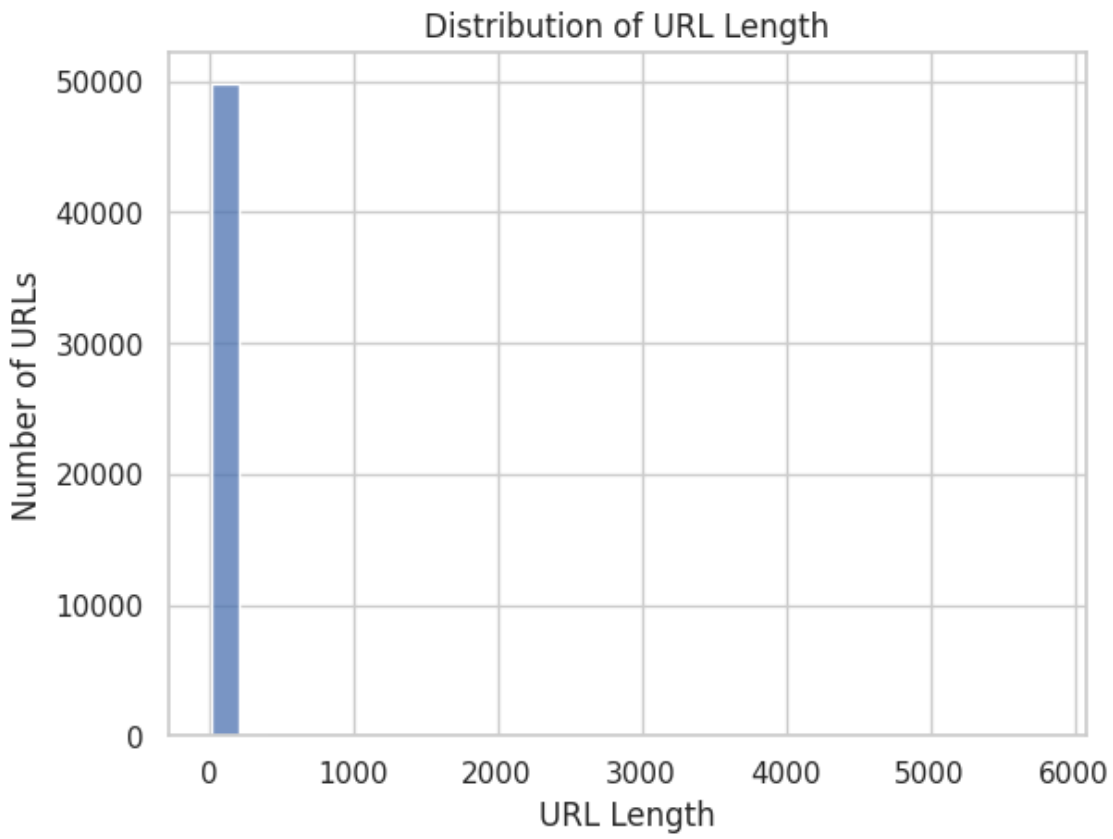
Key Findings

- Legitimate websites represent the majority of the dataset.
- Phishing websites account for a significant portion (41.25%).
- The class imbalance should be considered when developing phishing detection models.

Key Insight: The sample contains 20,625 phishing (41.25%) and 29,374 legitimate (58.75%) records.

Business Insight: Use class-aware metrics and stratified evaluation because the two classes are not equally represented.

Objective: To examine the distribution and extreme values of URL length.



URL length distribution

Interpretation: The distribution shows that most URLs have shorter lengths, while a smaller number of URLs are unusually long, resulting in a right-skewed distribution.

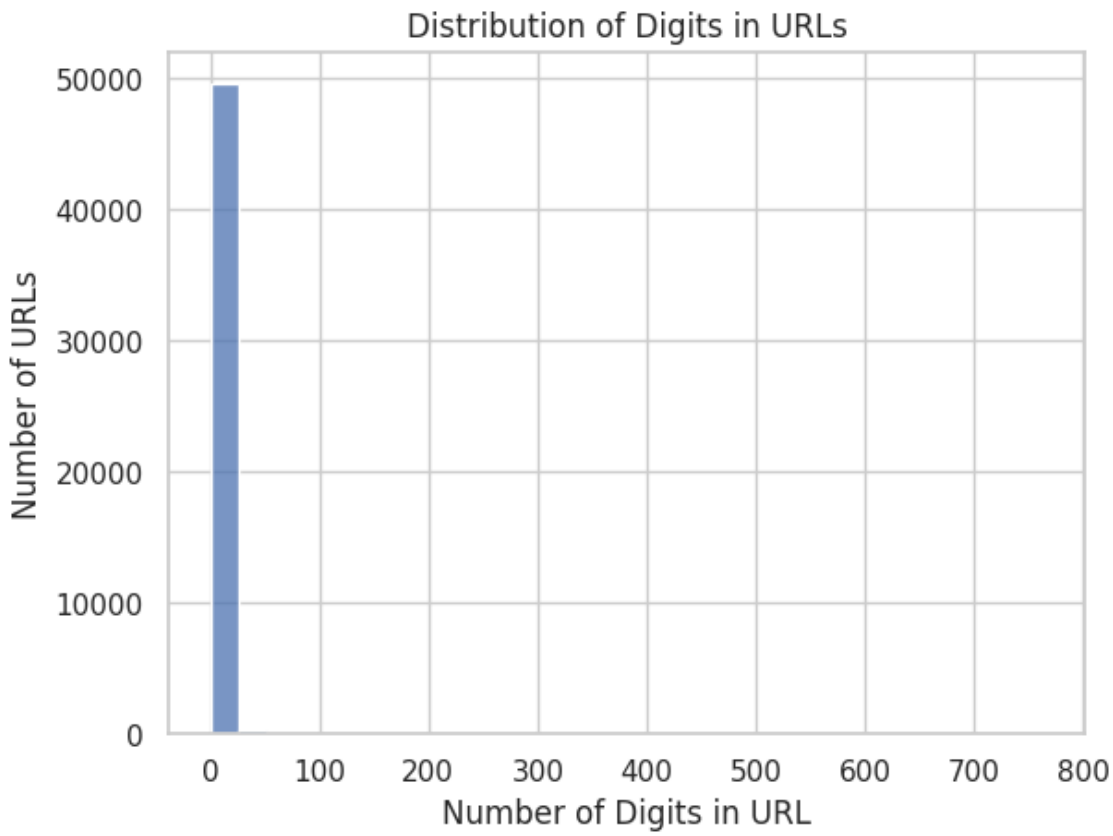
Key Findings

- Most URLs are concentrated in the shorter length range.
- A few URLs have extremely long lengths.
- URL length can be a useful feature for identifying unusual website patterns.
- Comparing URL lengths between phishing and legitimate websites may help identify suspicious URLs.

Key Insight: URL lengths are right-skewed, with relatively few exceptionally long URLs.

Business Insight: Investigate unusually long URLs as a risk signal, not as standalone proof of phishing.

Objective: To examine the distribution of numeric characters in URLs.



Digits in URLs distribution

Interpretation: The distribution shows that most URLs contain very few digits, while a smaller number contain many digits, resulting in a right-skewed distribution.

Key Findings

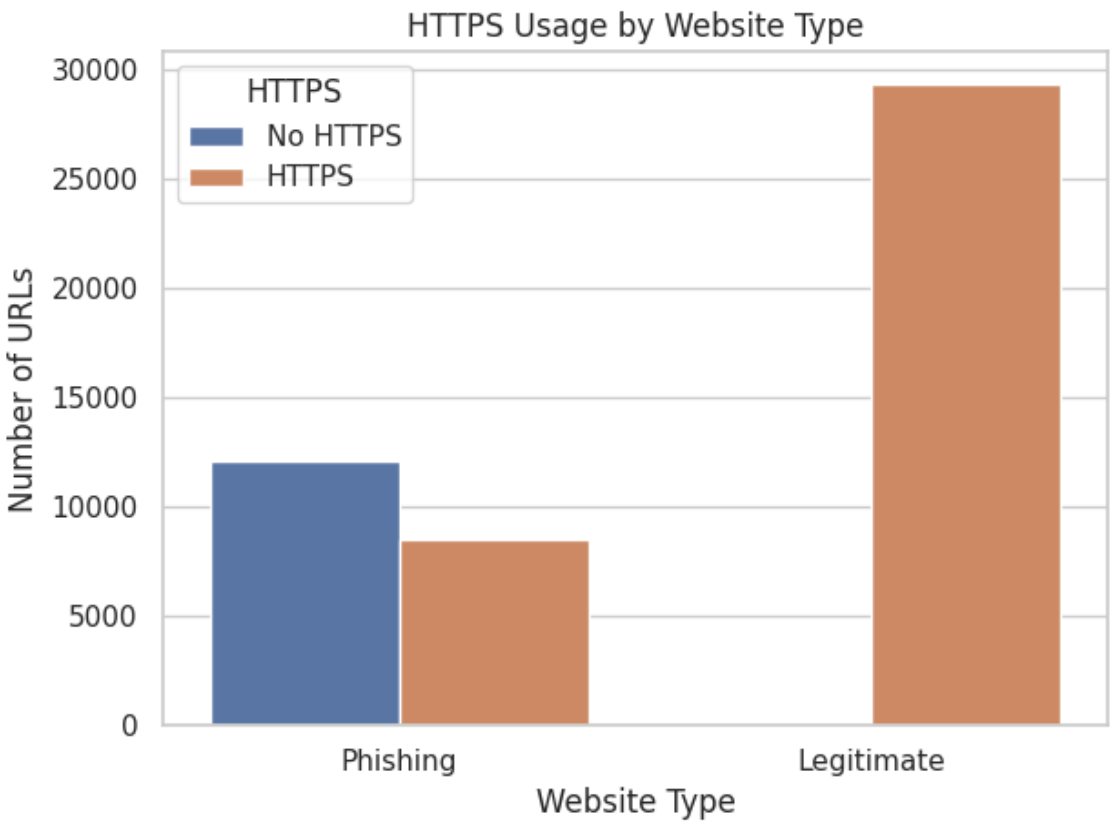
- Most URLs contain fewer digits.
- Some URLs contain unusually high numbers of digits.
- Digit count may help identify unusual URL patterns.
- Comparing digit counts between phishing and legitimate websites may help identify suspicious URLs.

Key Insight: Most URLs contain few digits, while a smaller group contains substantially more.

Business Insight: Combine digit counts with other indicators in URL screening rules.

11.2 Bivariate Analysis

Objective: To compare HTTPS adoption between phishing and legitimate websites.



HTTPS usage by website label

Interpretation: The chart compares HTTPS usage between phishing and legitimate websites.

Legitimate websites show higher HTTPS adoption, while phishing websites include both HTTPS and non-HTTPS URLs.

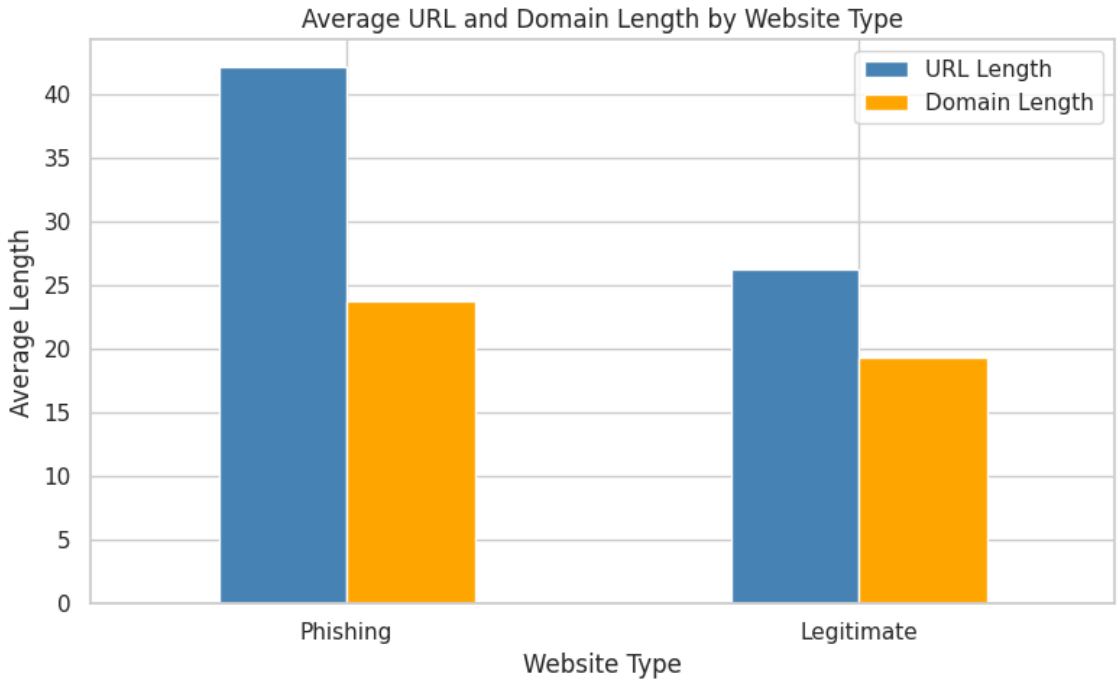
Key Findings

- HTTPS usage differs between phishing and legitimate websites.
- Some phishing websites also use HTTPS.
- HTTPS alone cannot confirm whether a website is safe.
- Combining HTTPS with other URL and website features may improve phishing detection.

Key Insight: HTTPS presence differs across the two website classes.

Business Insight: Do not treat HTTPS encryption alone as evidence that a website is safe.

Objective: To compare average URL and domain lengths by website class.



Average URL and domain length by label

Interpretation: The analysis shows that phishing websites have longer average URL and domain lengths compared to legitimate websites.

The average URL length is 42.14 for phishing websites and 26.28 for legitimate websites. Similarly, the average domain length is 23.70 for phishing websites and 19.28 for legitimate websites.

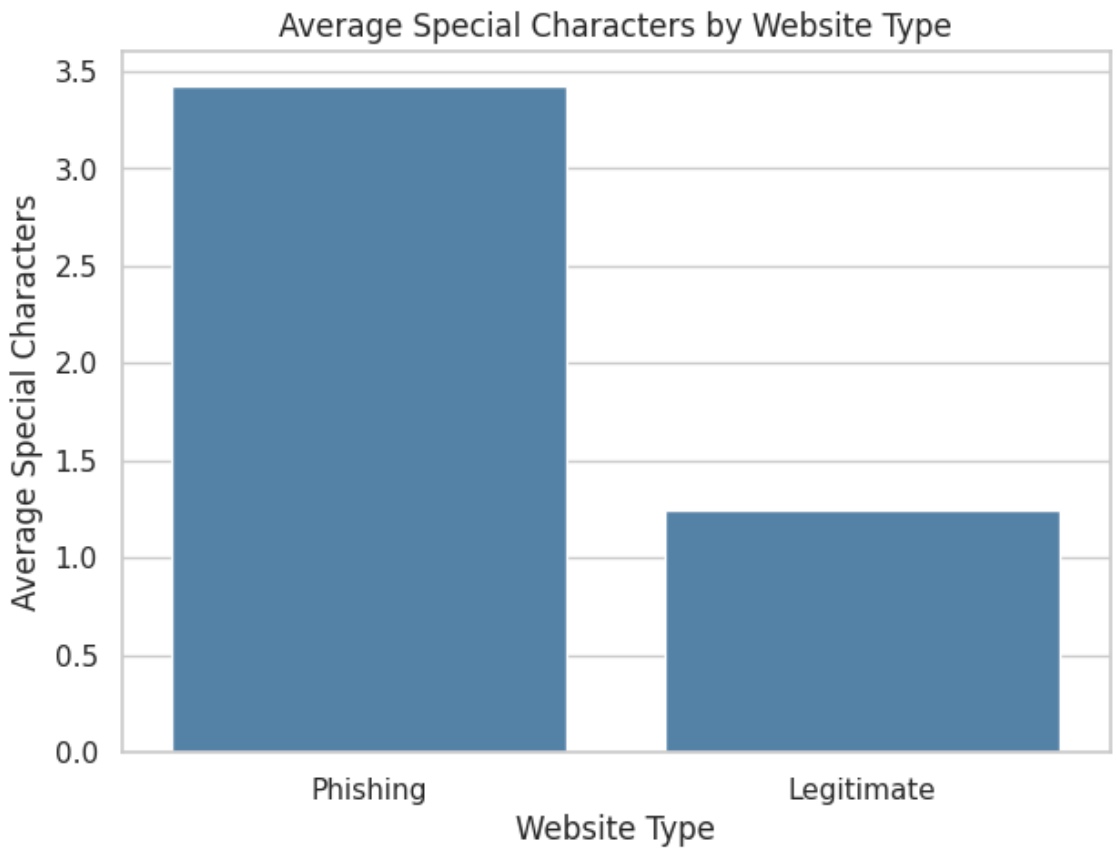
Key Findings

- Phishing websites have longer URLs on average.
- Phishing websites have slightly longer domain names.
- URL and domain lengths may help identify suspicious website patterns.
- These features can support phishing detection when combined with other security indicators.

Key Insight: Phishing URLs average 42.14 characters versus 26.28 for legitimate URLs; phishing domains average 23.70 versus 19.28.

Business Insight: Prioritize unusually long URLs and domains for additional investigation, using multiple indicators.

Objective: To compare special-character usage between website classes.



Average special-character counts by label

Interpretation: The chart shows that phishing URLs contain more special characters than legitimate URLs.

Phishing URLs have an average of 3.43 special characters, while legitimate URLs have an average of 1.24.

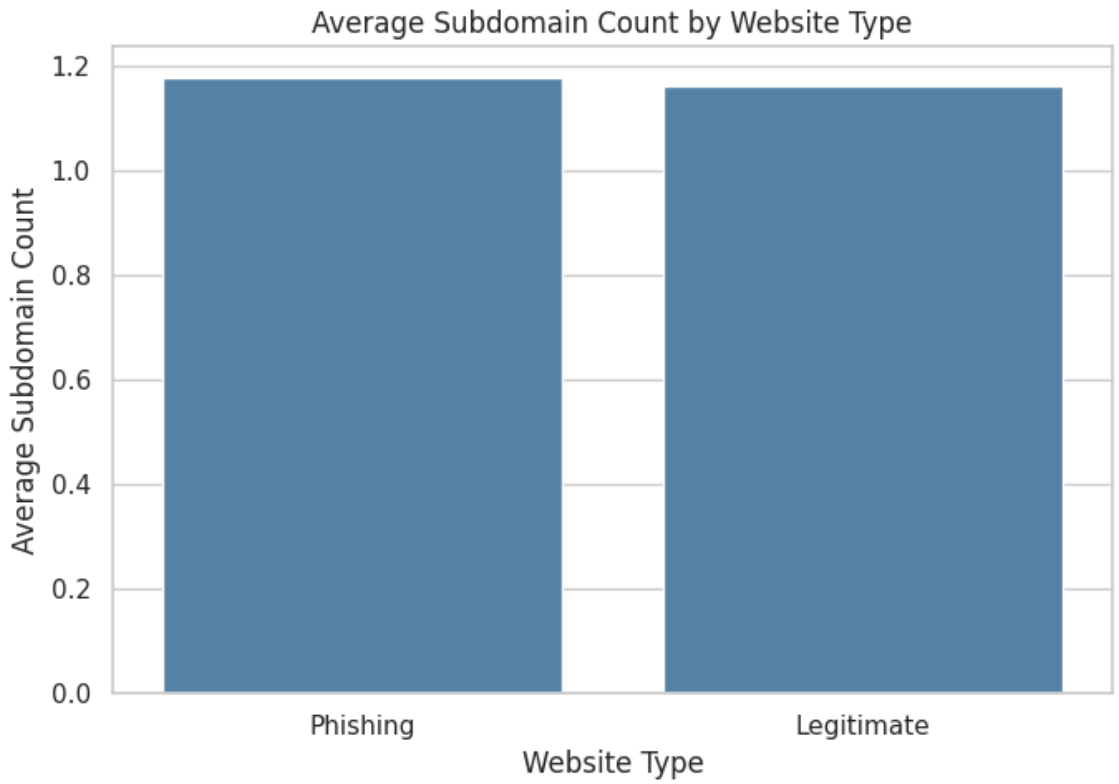
Key Findings

- Phishing URLs contain more special characters on average.
- Special characters may help identify suspicious URLs.
- Special characters alone cannot confirm a phishing website.
- Combining special character counts with other URL features may support phishing detection.

Key Insight: Phishing URLs contain 3.43 special characters on average, compared with 1.24 for legitimate URLs.

Business Insight: Include special-character frequency as one feature in phishing risk assessment.

Objective: To compare average subdomain counts across phishing and legitimate websites.



Average subdomain count by label

Interpretation: The statistical analysis and bar chart show that phishing and legitimate websites have nearly the same average number of subdomains.

Phishing websites have an average of 1.18 subdomains, while legitimate websites have an average of 1.16.

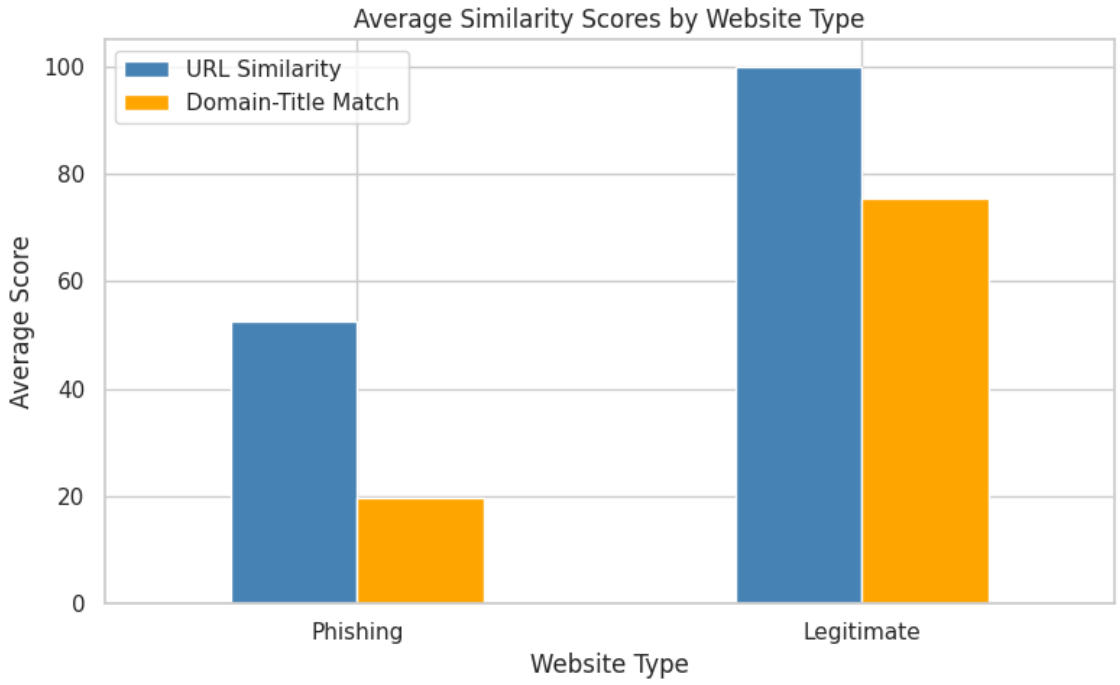
Key Findings

- Phishing websites have a slightly higher average subdomain count.
- The difference between both website types is very small.
- Subdomain count alone is not a strong indicator of phishing websites.
- Subdomain count may be more useful when combined with other URL and website features.

Key Insight: The average number of subdomains is similar across the two classes in the analyzed sample.

Business Insight: Do not rely on subdomain count alone to distinguish phishing from legitimate websites.

Objective: To assess URL similarity and website-title matching by class.



Average URL similarity and title-matching scores

Interpretation: The analysis shows that legitimate websites have higher URL similarity and domain-title matching scores compared to phishing websites.

The average URL similarity score is 52.69 for phishing websites and 100 for legitimate websites.

The average domain-title matching score is 19.76 for phishing websites and 75.44 for legitimate websites.

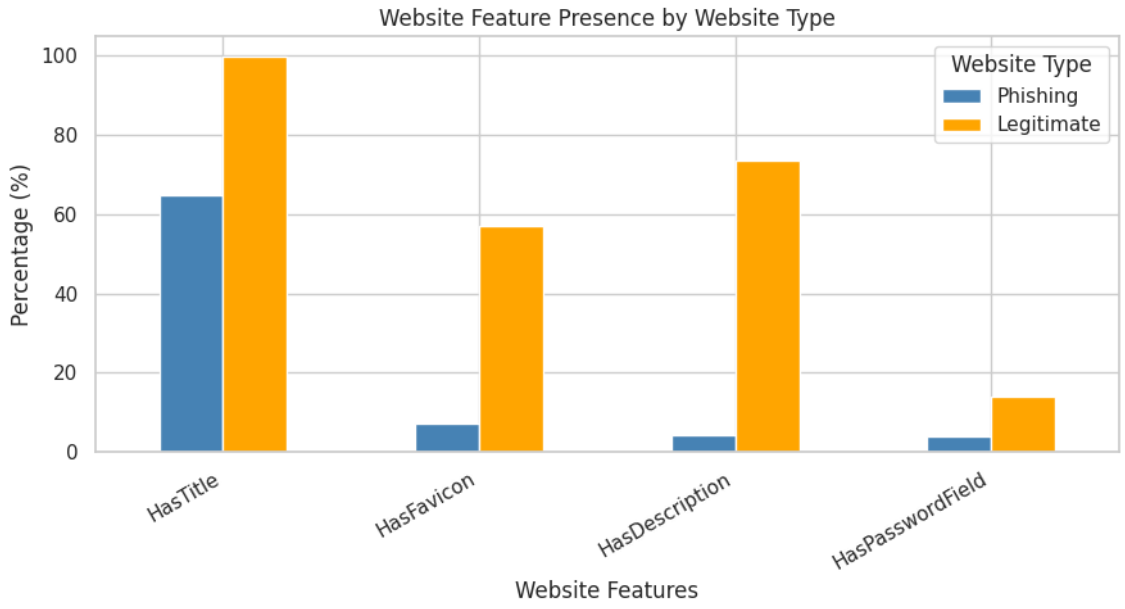
Key Findings

- Legitimate websites have higher URL similarity scores.
- Legitimate websites have stronger domain-title matching scores.
- These features may help distinguish phishing websites from legitimate websites.
- Lower similarity and domain-title matching scores may indicate suspicious website characteristics.
- Combining these features with other URL and security indicators may support phishing detection.

Key Insight: DomainTitleMatchScore averages 19.76 for phishing versus 75.44 for legitimate sites; URLTitleMatchScore averages 23.71 versus 75.44.

Business Insight: Low title-matching scores can support investigation, but feature availability and leakage risk should be verified.

Objective: To compare the prevalence of website titles, favicons and descriptions.



Website feature presence by label

Interpretation: The analysis shows that legitimate websites have a higher presence of titles, favicons, descriptions, and password fields compared to phishing websites.

Website titles are present in 99.87% of legitimate websites and 64.80% of phishing websites.

Website descriptions are present in 73.65% of legitimate websites and only 4.22% of phishing websites.

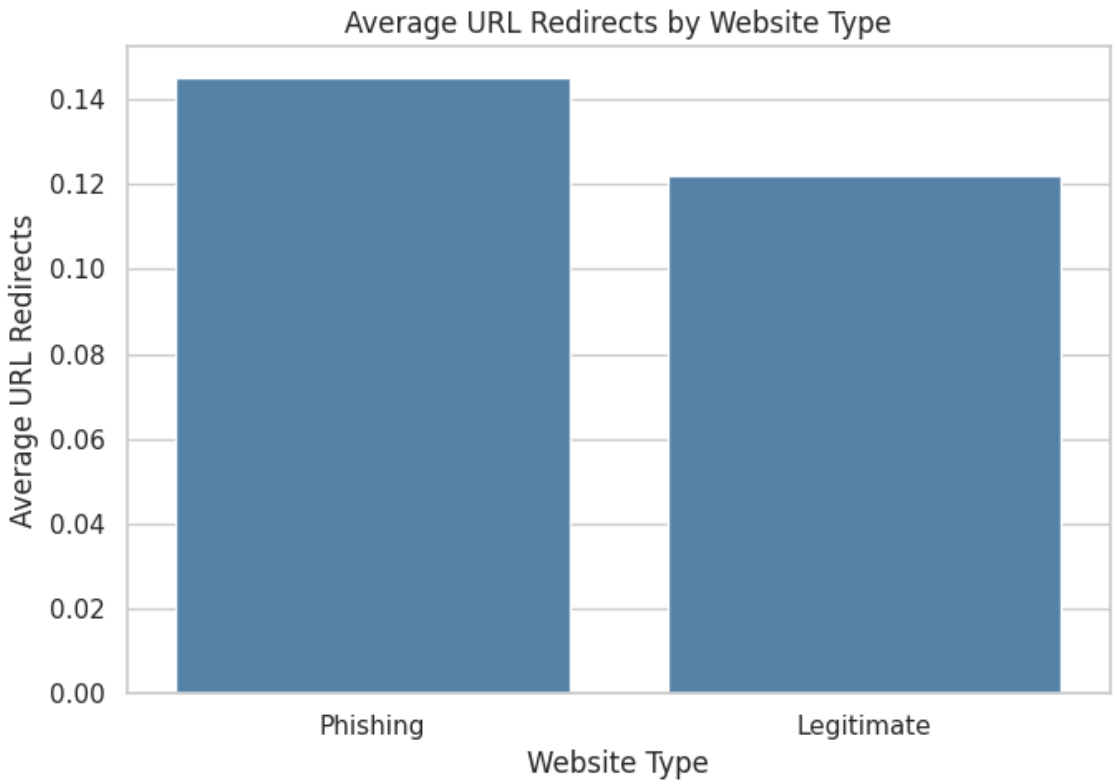
Key Findings

- Legitimate websites have higher presence rates for all four analyzed website features.
- Website titles and descriptions show noticeable differences between phishing and legitimate websites.
- Favicons are also more common among legitimate websites.
- The presence of password fields alone does not indicate phishing.
- Combining these website features with other URL characteristics may support phishing detection.

Key Insight: Titles appear on 64.80% of phishing versus 99.87% of legitimate sites; favicons on 7.25% versus 56.96%; descriptions on 4.22% versus 73.65%.

Business Insight: Missing common website elements can help flag pages for review, but are not definitive indicators.

Objective: To compare URL redirect behavior across website classes.



Average redirects by website label

Interpretation: The statistical analysis compares HTTPS usage, IP-based domains, and URL redirects between phishing and legitimate websites.

The bar chart shows that phishing websites have an average of 0.15 URL redirects, while legitimate websites have an average of 0.12.

Key Findings

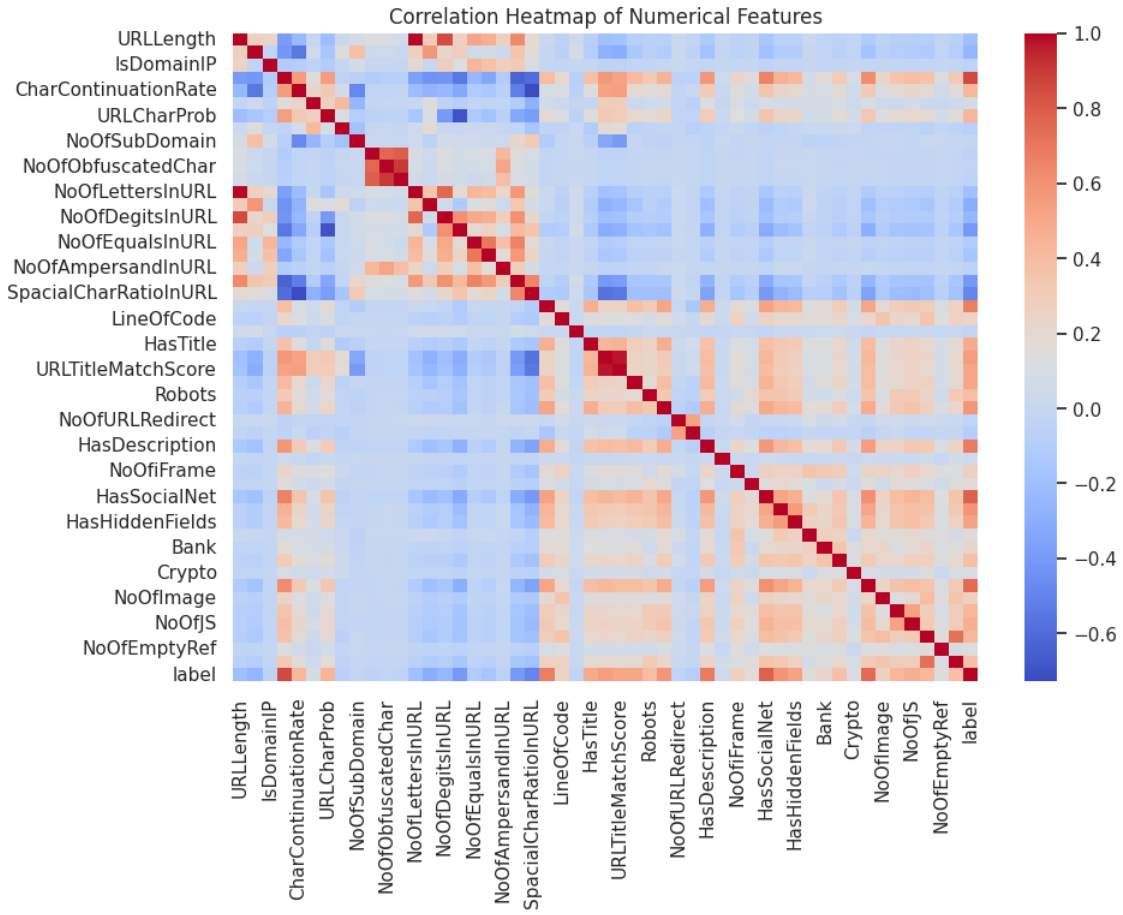
- Phishing websites have slightly more URL redirects on average.
- The difference in average redirects is very small.
- HTTPS usage, IP-based domains, and URL redirects provide useful information about website characteristics.
- These indicators alone cannot confirm whether a website is phishing.
- Combining multiple URL and security features may support more effective phishing detection.

Key Insight: The visualization compares average redirect counts by class; exact values should be read from the plotted results.

Business Insight: Review redirect patterns together with URL and domain signals when prioritizing investigations.

11.3 Multivariate Analysis

Objective: To identify relationships among numerical website and URL features.



Correlation heatmap of numerical features

Interpretation: The correlation heatmap shows the relationships among numerical URL and website features.

Positive correlations indicate that two features tend to increase together, while negative correlations indicate that one feature tends to decrease as the other increases.

Key Findings

- The heatmap helps identify relationships between numerical features.
- Strongly correlated features may contain similar information.
- Understanding these relationships can support feature selection for phishing detection.
- Correlation does not necessarily indicate a cause-and-effect relationship.

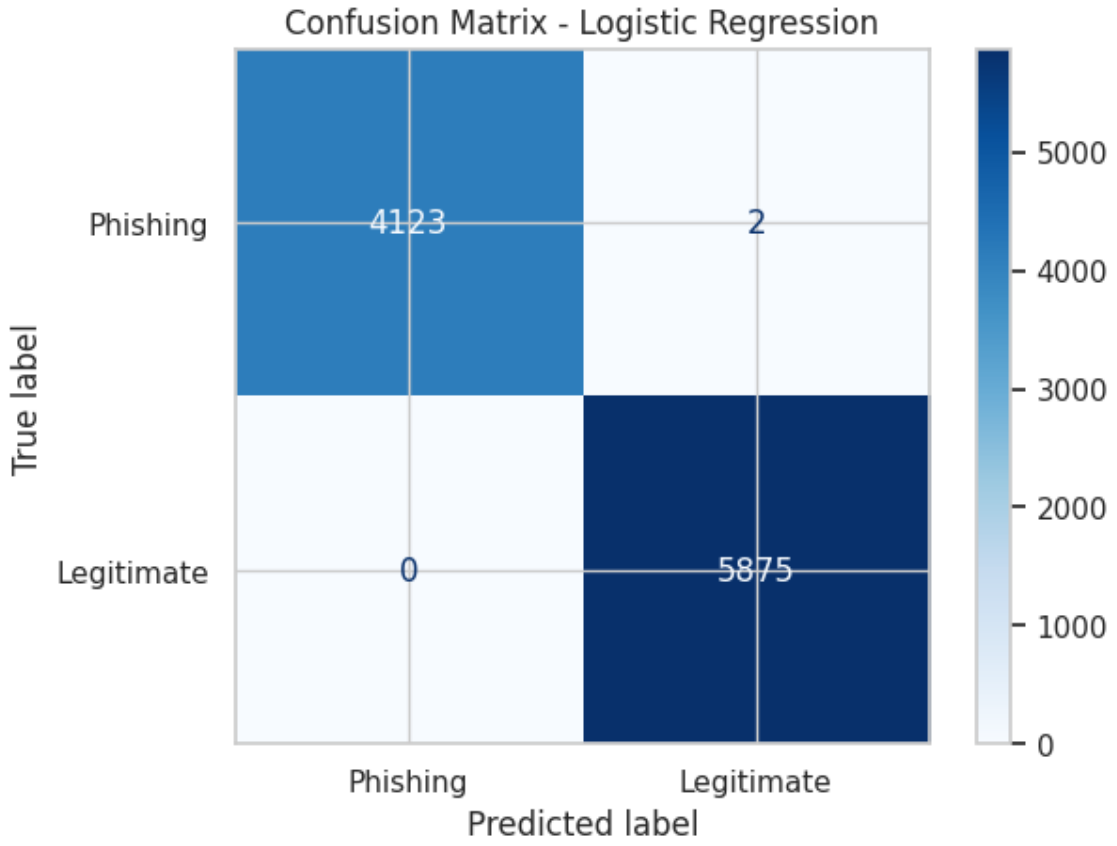
Key Insight: The heatmap visualizes pairwise feature correlations; correlation alone does not establish causation or data leakage.

Business Insight: Review redundant or unusually predictive features before deployment and validate on independent data.

11.4 Predictive Model Evaluation

A StandardScaler and Logistic Regression pipeline was trained on an 80% stratified training split and evaluated on a 20% test split.

Objective: To evaluate phishing classification errors on held-out test records.



Logistic Regression confusion matrix

Interpretation: On the 10,000-record test split, the model correctly classified 9,998 records and misclassified 2 phishing records as legitimate. Results describe this sampled test split, not independent real-world performance.

Key Insight: The model classified 9,998 of 10,000 test records correctly, missing 2 phishing websites and producing no false phishing alarms in this split.

Business Insight: Prioritize phishing recall and missed-phishing cases; confirm performance on independent, newer domains.

Metric	Result
Original accuracy	99.98%
Phishing precision	100.00%
Phishing recall	99.95%
Phishing F1-score	99.98%
Validation accuracy (four features removed)	99.92%
Validation phishing recall	99.83%

Feature sensitivity analysis excluded URLSimilarityIndex, TLDLegitimateProb, DomainTitleMatchScore and URLTitleMatchScore. Strong accuracy persisted, but this does not eliminate possible data leakage or prove performance on new domains.

12. Summary of Findings

1. URL and Domain Length

Phishing websites had a higher average URL length (42.14) compared to legitimate websites (26.28).

The average domain length was also higher for phishing websites (23.70) than legitimate websites (19.28).

Insight: Longer URLs and domains were more common among phishing websites in the analyzed dataset.

2. Special Characters in URLs

Phishing websites contained an average of 3.43 special characters, compared to 1.24 for legitimate websites.

Insight: A higher number of special characters may be a useful indicator of potentially suspicious URLs.

3. Domain and Website Title Matching

Legitimate websites had higher average DomainTitleMatchScore (75.44) and URLTitleMatchScore (75.44).

Phishing websites recorded lower averages of 19.76 and 23.71, respectively.

Insight: Lower matching scores were associated with phishing websites in this dataset.

4. Website Features

Legitimate websites showed a higher presence of common website features:

- Website Title: 99.87% legitimate vs. 64.80% phishing.
- Favicon: 56.96% legitimate vs. 7.25% phishing.
- Description: 73.65% legitimate vs. 4.22% phishing.

Insight: The presence of common website features differed substantially between phishing and legitimate websites.

5. Predictive Model Performance

The original Logistic Regression model achieved 99.98% accuracy and 99.95% phishing recall on the test dataset.

After excluding four selected features, the validation model achieved 99.92% accuracy and 99.83% phishing recall.

Insight: The model demonstrated strong performance on the sampled dataset, but additional independent testing and feature leakage checks are required before real-world deployment.

13. Types of Analysis

1. Descriptive Analysis — What Happened?

Analyzed 49,999 website records to understand the distribution of phishing and legitimate websites.

- Phishing Websites: 20,625 (41.25%)
- Legitimate Websites: 29,374 (58.75%)

Examined URL length, domain characteristics, HTTPS usage, and website features using summary statistics and visualizations.

2. Diagnostic Analysis — What Factors Are Associated?

Compared phishing and legitimate websites to identify characteristics associated with phishing.

- Phishing websites had longer URLs and more special characters on average.
- Lower domain-title and URL-title matching scores were associated with phishing websites.
- Website features such as favicons and descriptions showed noticeable differences between the two categories.

These findings identify associations but do not establish the causes of phishing.

3. Predictive Analysis — What Might Happen?

Developed a Logistic Regression classification model to predict whether a website is phishing or legitimate.

- Original Model Accuracy: 99.98%
- Original Model Phishing Recall: 99.95%
- Validation Model Accuracy: 99.92%
- Validation Model Phishing Recall: 99.83%

The model demonstrated strong classification performance on the sampled test data. Further independent validation is necessary to assess its ability to identify new phishing websites.

4. Prescriptive Analysis — What Should Be Done?

Based on the analysis, the following actions are recommended:

- Evaluate multiple URL and website characteristics rather than relying on a single indicator.
- Treat unusual URL structures and low domain-title matching scores as potential warning signals.
- Avoid assuming that HTTPS alone guarantees website legitimacy.
- Conduct additional model validation before considering deployment in real-world phishing detection systems.

These are recommendations derived from the analysis, not outcomes of a deployed detection system.

14. Future Enhancements and Suggestions

The following improvements can be considered to enhance the accuracy, reliability, and practical applicability of the phishing website detection model.

1. Expand the Dataset

- Include more recent phishing and legitimate website records.
- Use datasets from multiple sources to improve diversity and generalization.

2. Compare Machine Learning Models

- Evaluate additional algorithms such as Random Forest, Decision Tree, and XGBoost.
- Compare models using accuracy, precision, recall, F1-score, and confusion matrices.

3. Improve Feature Selection

- Investigate potentially misleading or label-dependent features.
- Apply feature selection techniques to identify the most useful predictors.
- Verify that selected features can be calculated for new websites during prediction.

4. Strengthen Model Validation

- Test the model using independent datasets and newly collected URLs.
- Evaluate performance on previously unseen domains.
- Investigate potential data leakage and monitor false-negative predictions.

5. Develop a Phishing Detection Application

- Create a simple web application using Streamlit.
- Allow users to enter a website URL and receive a phishing risk prediction.
- Add appropriate warnings and explain that predictions are not guarantees of website safety.

Expected Outcome

These future enhancements could improve model generalization, support more reliable phishing detection, and help transform the current analytical project into a practical cybersecurity application.

15. Conclusion

This project successfully analyzed phishing and legitimate websites using Python to understand the characteristics associated with phishing websites.

Through data cleaning, exploratory data analysis, visualization, and predictive modeling, the project identified useful patterns and demonstrated the potential of machine learning in phishing website detection.

Overall, the project achieved its objectives and highlighted how data-driven techniques can support website security analysis. Further testing with real-world data can help improve the reliability of the model.

Additional Recommendations and Decision Support

Recommendations

7. Analyze URL Characteristics

- Examine URL length, domain length, and special characters to identify potentially suspicious websites.
- Avoid classifying websites based on URL length alone.

8. Evaluate Domain and Website Features

- Consider domain-title matching, URL-title matching, and the presence of website descriptions and favicons.
- Use multiple indicators together to support phishing detection.

9. Avoid Relying Only on HTTPS

- HTTPS indicates an encrypted connection but does not guarantee that a website is legitimate.
- Combine HTTPS information with other URL and website characteristics.

10. Improve Model Reliability

- Validate the model using independent and more recent website data.
- Investigate potential feature leakage and verify that all selected features are available during prediction.

- Monitor false negatives because missed phishing websites may expose users to security risks.

Decision Support

The analysis can support cybersecurity decision-making by:

- Highlighting website characteristics associated with phishing.
- Helping prioritize potentially suspicious URLs for further investigation.
- Providing a baseline classification model for future development.
- Supporting the design of a multi-feature phishing detection approach.

Expected Benefits

These recommendations could help improve phishing risk assessment and guide the development of more reliable detection systems.

However, the proposed benefits have not been evaluated through real-world deployment.